

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Debugging & Optimisation Task (Low)
Assessment Tool Weightage: 10%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.	BL2 – Understand
2	Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:	BL2 – Understand
3	Correct a faulty SELECT query where column names are misspelled or improperly referenced.	BL1 – Remember
4	Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.	BL3 – Apply
5	Identify and correct a subquery error where multiple values are returned instead of a single value.	BL2 – Understand
6	Debug a VIEW creation statement that contains invalid column references or syntax errors.	BL1 – Remember
7	Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.	BL3 – Apply
8	Identify violations of integrity constraints in a given table and correct them.	BL2 – Understand

9	Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.	BL3 – Apply
10	Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.	BL3 – Apply

Code of Conduct:

I Geferlin Boffy (192571080) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1) Create Table Debugging

Incorrect code:

```
mysql> CREATE TABLE Employee
-> (
-> EmpID INT
-> EmpName VARCHAR(30),
-> Salary INT
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check
the manual that corresponds to your MySQL server version for the
right syntax to use near 'EmpName VARCHAR(30),
Salary INT
)' at line 4
```

Error Identified:

- The CREATE TABLE statement contained syntax errors and missing constraints such as PRIMARY KEY and NOT NULL.

Correct code:

```
mysql> CREATE TABLE Employee
-> (
-> EmpID INT PRIMARY KEY,
-> EmpName VARCHAR(30) NOT NULL,
-> Salary INT
-> );
Query OK, 0 rows affected (0.02 sec)
```

The syntax errors were fixed and the required constraints were added. The table was created successfully.

```
mysql> DESC Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(30)	NO		NULL	
Salary	int	YES		NULL	

3 rows in set (0.02 sec)

2) Aggregate Function Debugging

```
mysql> SELECT Salary,COUNT(*)
```

```
-> FROM Employee;
```

```
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'co2_at3.Employee.Salary'; this is incompatible with sql_mode=only_full_group_by
```

```
mysql> SELECT Salary,  
-> COUNT(*) AS Total  
-> FROM Employee  
-> GROUP BY Salary;
```

Salary	Total
25000	1
35000	1
28000	1

```
3 rows in set (0.00 sec)
```

3) Column Name Correction

```
mysql> SELECT StudntName
```

```
-> FROM Student;
```

```
ERROR 1054 (42S22): Unknown column 'StudntName' in 'field list'
```

```
mysql> |
```

Incorrect Query:

The query contains a misspelled column name, causing an unknown column error.

```
mysql> SELECT StudentName  
-> FROM Student;
```

StudentName
Arun
Priya
Rahul
Divya
Karthik

5 rows in set (0.00 sec)

Corrected Query:

The column name was corrected, and the query executed successfully.

4) JOIN Query Correction

```
mysql> SELECT *  
-> FROM Student  
-> JOIN Department;
```

RegNo	StudentName	Age	DeptID	DeptID	DeptName
101	Arun	20	1	3	EEE
101	Arun	20	1	2	ECE
101	Arun	20	1	1	CSE
102	Priya	19	2	3	EEE
102	Priya	19	2	2	ECE
102	Priya	19	2	1	CSE
103	Rahul	21	1	3	EEE
103	Rahul	21	1	2	ECE
103	Rahul	21	1	1	CSE
104	Divya	20	3	3	EEE
104	Divya	20	3	2	ECE
104	Divya	20	3	1	CSE
105	Karthik	22	2	3	EEE
105	Karthik	22	2	2	ECE
105	Karthik	22	2	1	CSE

15 rows in set (0.00 sec)

Incorrect Query:

The JOIN statement is missing the join condition, resulting in an incorrect query.

```
mysql> SELECT Student.StudentName,
-> Department.DeptName
-> FROM Student
-> JOIN Department
-> ON Student.DeptID=Department.DeptID;
```

StudentName	DeptName
Arun	CSE
Rahul	CSE
Priya	ECE
Karthik	ECE
Divya	EEE

5 rows in set (0.00 sec)

Corrected Query:

The correct ON condition was added, and the tables were joined successfully

5) Subquery Correction

```
mysql> SELECT *
-> FROM Student
-> WHERE DeptID=
-> (
-> SELECT DeptID
-> FROM Department
-> );
ERROR 1242 (21000): Subquery returns more than 1 row
```

Incorrect Query:

The subquery returns multiple rows where only a single value is expected.

```
mysql> SELECT *
-> FROM Student
-> WHERE DeptID IN
-> (
-> SELECT DeptID
-> FROM Department
-> );
```

RegNo	StudentName	Age	DeptID
101	Arun	20	1
103	Rahul	21	1
102	Priya	19	2
105	Karthik	22	2
104	Divya	20	3

5 rows in set (0.00 sec)

Corrected Query:

The IN operator was used to handle multiple values, and the query executed successfully.

6) VIEW Debugging

```
mysql> CREATE VIEW StudentView AS
-> SELECT StudentNam
-> FROM Student;
ERROR 1054 (42S22): Unknown column 'StudentNam' in 'field list'
```

Incorrect Query:

The VIEW creation failed because the view already existed (or contained an invalid reference).

```
mysql> SELECT * FROM StudentView;
+-----+
| StudentName |
+-----+
| Arun        |
| Priya       |
| Rahul       |
| Divya       |
| Karthik     |
+-----+
5 rows in set (0.00 sec)
```

Corrected Query:

The existing view was removed and recreated successfully with the correct SQL statement.

7) Query Optimization

Original Query:

The query contains redundant conditions, making it unnecessarily complex.

```
mysql> SELECT *
-> FROM Student
-> WHERE Age>18
-> AND Age>19;
+-----+-----+-----+-----+
| RegNo | StudentName | Age | DeptID |
+-----+-----+-----+-----+
| 101   | Arun        | 20  | 1       |
| 103   | Rahul       | 21  | 1       |
| 104   | Divya       | 20  | 3       |
| 105   | Karthik     | 22  | 2       |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Optimized Query:

The redundant condition was removed, producing the same result with a simpler and more efficient query.

```
mysql> SELECT *
-> FROM Student
-> WHERE Age>19;
+-----+-----+-----+-----+
| RegNo | StudentName | Age | DeptID |
+-----+-----+-----+-----+
| 101 | Arun | 20 | 1 |
| 103 | Rahul | 21 | 1 |
| 104 | Divya | 20 | 3 |
| 105 | Karthik | 22 | 2 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Same output, but the optimized query is simpler because it removes the redundant condition.

8) Integrity Constraint Correction

```
mysql> INSERT INTO Student2
-> VALUES
-> (101, NULL, 15);
ERROR 1048 (23000): Column 'StudentName' cannot be null
```

Incorrect Data:

The inserted data violates one or more integrity constraints, causing the insertion to fail.

```
mysql> INSERT INTO Student2
-> VALUES
-> (101, 'Meena', 19);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT *
-> FROM Student2;
+-----+-----+-----+
| RegNo | StudentName | Age |
+-----+-----+-----+
| 101 | Meena | 19 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

Corrected Data:

The data was modified to satisfy all integrity constraints and was inserted successfully.

9) 1NF / 2NF Normalization

Original relation containing repeating groups (violates 1NF).

```
mysql> SELECT * FROM StudentCourse;
+-----+-----+-----+-----+
| RegNo | StudentName | Course1 | Course2 |
+-----+-----+-----+-----+
| 101   | Arun        | DBMS    | Java    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Creation of normalized tables.

```
mysql> CREATE TABLE StudentNew
-> (
-> RegNo INT PRIMARY KEY,
-> StudentName VARCHAR(30)
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE Registration
-> (
-> RegNo INT,
-> CourseName VARCHAR(30),
-> PRIMARY KEY(RegNo,CourseName)
-> );
Query OK, 0 rows affected (0.01 sec)
```

Data inserted into normalized tables.

```
mysql> INSERT INTO StudentNew VALUES
-> (101,'Arun');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Registration VALUES
-> (101,'DBMS'),
-> (101,'Java');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

Normalized tables after removing repeating groups (1NF achieved)

```
mysql> SELECT * FROM StudentNew;
+-----+-----+
| RegNo | StudentName |
+-----+-----+
| 101   | Arun        |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM Registration;
+-----+-----+
| RegNo | CourseName |
+-----+-----+
| 101   | DBMS       |
| 101   | Java       |
+-----+-----+
2 rows in set (0.00 sec)
```

Original Relation:

The table contains repeating groups, violating the First Normal Form (1NF).

Below Normalized Tables Screenshot

Normalized Relation:

The relation was decomposed into separate tables, eliminating repeating groups and improving data organization.

10) 3NF Normalization

Creating the Department table to store department details separately.

```
mysql> CREATE TABLE DepartmentNew
-> (
-> DeptID INT PRIMARY KEY,
-> DeptName VARCHAR(30),
-> DeptPhone VARCHAR(15)
-> );
Query OK, 0 rows affected (0.02 sec)
```

Creating the Student table with a foreign key reference.

```
mysql> CREATE TABLE StudentFinal
-> (
-> RegNo INT PRIMARY KEY,
-> StudentName VARCHAR(30),
-> DeptID INT,
-> FOREIGN KEY(DeptID)
-> REFERENCES DepartmentNew(DeptID)
-> );
Query OK, 0 rows affected (0.03 sec)
```

Inserting data into the normalized tables.

```
mysql> INSERT INTO DepartmentNew VALUES
-> (1,'CSE','9876543210'),
-> (2,'ECE','9876500000');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO StudentFinal VALUES
-> (101,'Arun',1),
-> (102,'Priya',2);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

Normalized tables after applying Third Normal Form (3NF).

```
mysql> select * from studentfinal;
+-----+-----+-----+
| RegNo | StudentName | DeptID |
+-----+-----+-----+
| 101   | Arun        | 1      |
| 102   | Priya       | 2      |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select * from DepartmentNew;
+-----+-----+-----+
| DeptID | DeptName | DeptPhone |
+-----+-----+-----+
| 1      | CSE      | 9876543210 |
| 2      | ECE      | 9876500000 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```